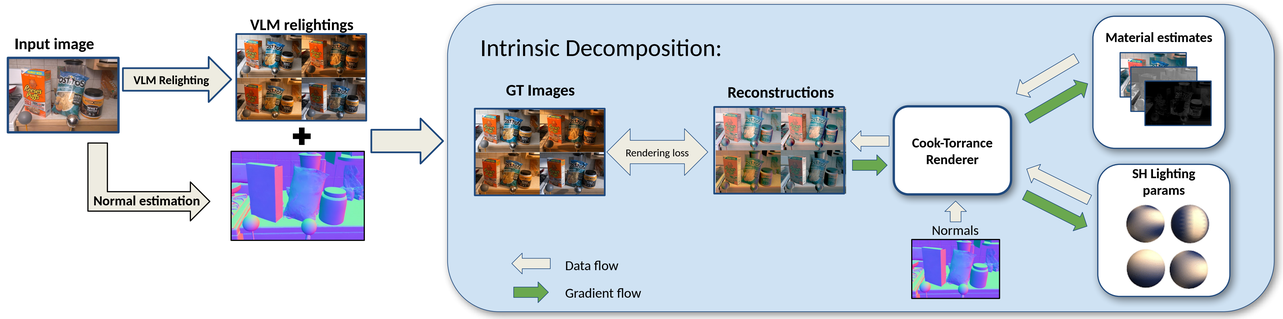
PBR Estimation with Diffusion-generated Relightings

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Motivation

- Existing PBR estimators use fine tuned diffusion models trained on synthetic indoor data → bad generalization.
- VLMs have good understanding of lighting. → Use VLMs prior knowledge to estimate material properties from single photo.

Inverse Rendering Pipeline:



Method

Input: Single RGB image **Output:** PBR material decomposition (albedo + roughness + metallic)



Optimization Priors:

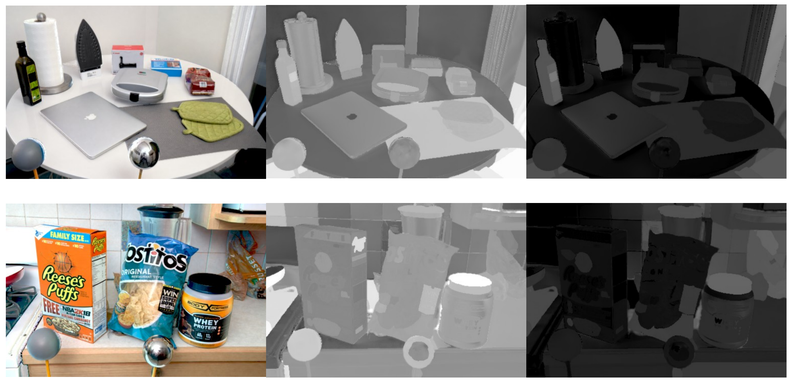
- Per object material similarity
- Metallic L1 prior
- Spatial smoothness prior

Ablations

Without per object material:



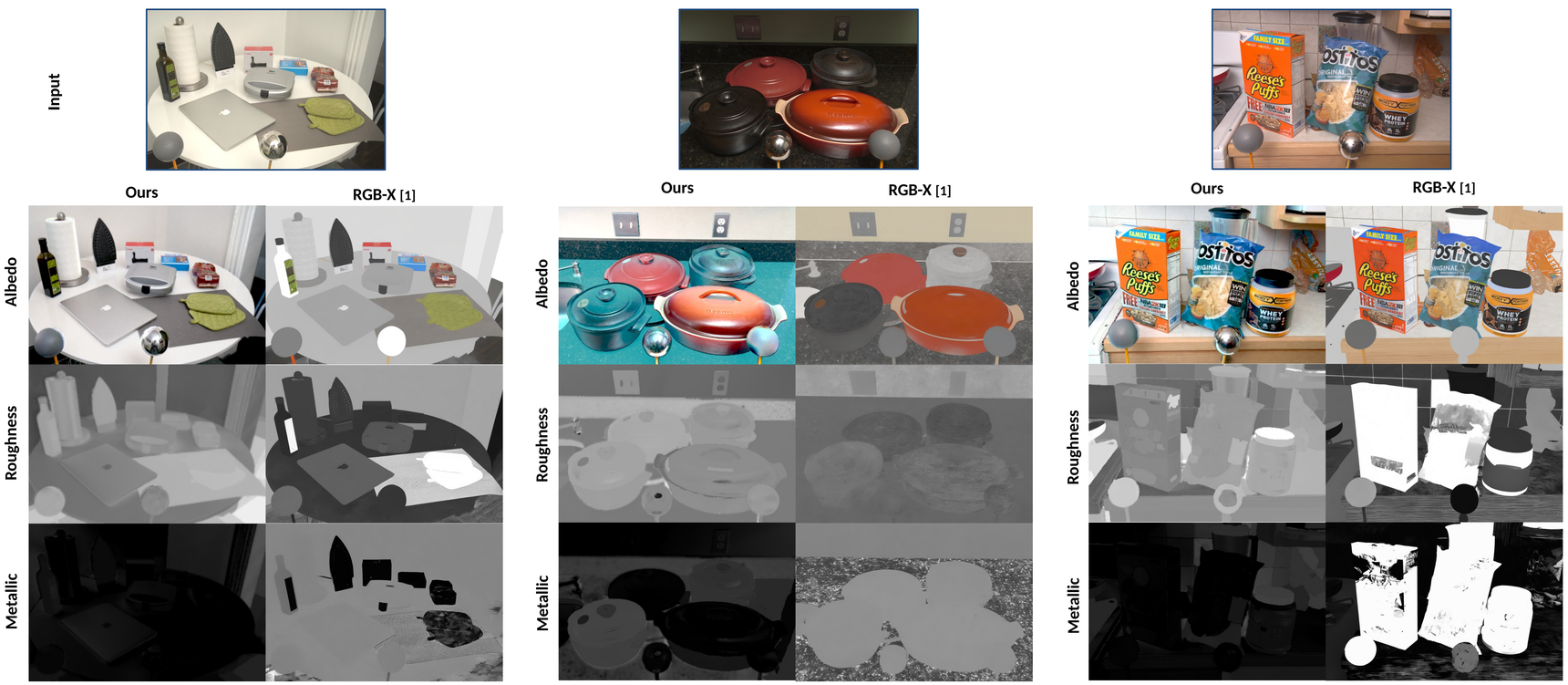
Without metallic L1:



No priors:



Results



[1] *Zeng et al., SIGGRAPH 2024 — RGB ↔ X*

Dataset: *Murmann et al., ICCV 2019 — MIT Multi-Illumination Dataset, https://data.csail.mit.edu/multilum/*